

the absence of shock-absorbing elements in the real hardware, as well as the limitations in joint torques and velocities, prevent us from demonstrating these gaits in an experimental setting.

- Composite actions combine two or more actions into one standalone entity. A composite action extends the basic motion of one of the actions presented so far with additional subtasks, for example, by adding arm and hand motions during walking for a loco-manipulation task. In demonstration A, a composite action is used for the cushion grasping motion, combining arm motions (Move-Limb actions for the left and right arms) with a vertical CoM adjustment (Shift-CoM action). The commanded CoM vertical shift ensures that the robot's hips maintain a relatively constant height above ground during the arms' upward motion.

The complete sequence of actions corresponding to the high-level command performed in demonstration A is given in Table 1. In general, the action parameters used in this demonstration are fixed because of the predefined environment (lean forward 18°, two stair steps). Ideally, these parameters could be captured or computed according to an advanced perception algorithm that interprets the environment to acquire the corresponding parameters for each action template. Note also that while the sequence of actions is predefined, the locations of the waypoints 2, 3, and 4 are specified relative to the tag poses, which are unknown at the beginning of the demonstration. Therefore, the concrete actions composing the action sequence can only be instantiated when the tag locations are determined. During the localization procedure, while the vision module reorients the camera by moving the robot's head and the tag detection algorithm is executed, the action sequencer is in idle mode, waiting for the localization result. This situation occurs at waypoint 1 (start pose) for tag 1 localization and at waypoint 2 for tag 2, after executing the first Walk-to-Waypoint action. Even though no explicit action is generated, this situation is interpreted by the plan assembler and the reference trajectory generator as an instruction to maintain the current reference whole-body task while balancing. The same behavior is implemented for demonstration C (see Figure 4), whenever there is no operator input given through the joystick controller. We discuss these cases in more detail below.

FOOTSTEP PLANNER

The footstep planning process is initiated by the action sequencer whenever a locomotion action is generated. Given the robot's 2D pose at the end of the previous action and the goal pose of the current locomotion action, the footstep planner generates a sequence of contacts with alternating left and right footsteps connecting the two poses [see Figure 6(b)]. If the previous action is also a locomotion action, the first footstep is chosen such that it naturally continues the existing sequence; i.e., if the previous sequence ends with the left foot, then the first footstep of the current sequence is taken with the right foot, and vice versa. For other types of preceding actions or in the initial pose, the first footstep is determined

based on the motion type [see Figure 6(a)]: forward and backward walking start with the right foot, and lateral walking and turning start with the foot corresponding to the direction of motion (right foot if walking to the right, left foot for left turns, etc.). A similar algorithm is employed when the actions are generated online via a joystick interface: during walking, the footstep planner generates alternating footsteps; when no commands are given, the robot stops and maintains balance in a standing pose; and finally, when commands are resumed, the initial footstep is determined by the direction of motion.

The parameters guiding the footstep planning process are the maximum step length on the x - and y -axes and the maximum turning angle during one step. To prevent collisions between the feet, the footstep planner avoids footstep placements that would cause the swing foot to cross the body's sagittal plane. Furthermore, potential knee collisions are averted by prohibiting the inward turning of the footsteps. The effect of this strategy can be seen in Figure 6(c), where the walking trajectory curves to the right, as seen in the local coordinates. Here only the right footsteps contribute to the turning motion as they correspond to outward turns of the foot; in contrast, the left footsteps maintain the same orientation as the preceding right footsteps.

The result of the footstep planning process is a chain of contacts, where each contact is linked to the preceding one by a constraint formulated as a footstep placement bounding area [see Figure 6(c)]. This information is used by the footstep adjustment module, which can make changes to the contact positions in response to unrecoverable tracking errors reported by the whole-body controller. For example, when adjusting the constrained contact labeled i , the adjustment offset is limited by the requirement that the footstep remains inside the bounding area associated with the reference contact $i - 1$, marked in gray in Figure 6(c). Moreover, as the contact i is

TABLE 1. The action sequence for cushion retrieval with vision-based locomotion (demonstration A).

ACTION	PARAMETER/COMMENT
Walk-to-Waypoint	Waypoint 2 (relative to tag 1)
Walk-to-Waypoint	Waypoint 3 (relative to tag 2)
Turn-in-Place	Align yaw angle with tag 2
Walk-to-Waypoint	Waypoint 4 (lateral walking)
Change-Orientation	Lean forward 18°
Climb-Stairs	Two stair steps
Change-Orientation	Stand upright
Grasp-Cushion	Composite action
Descend-Stairs	Two stair steps
Lower-Arms	Composite action
Walk-to-Waypoint	Waypoint 3 (lateral walking)
Walk-to-Waypoint	Waypoint 1 (start pose)
Release-Cushion	Composite action